Compound 2D Shapes

Name _____ Date _____ Calculator allowed Version 1

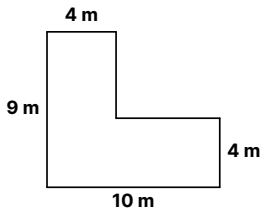
KEY WORDS composite shape · perimeter · area · rectangle · triangle · trapezium · semicircle

A Shapes made from rectangles ●○○ WORK IN YOUR BOOK

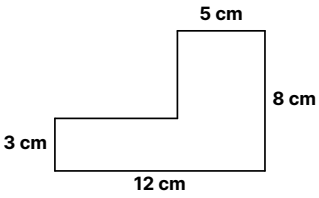
Remember: Find the missing side lengths first. For the area, split the shape into rectangles.

1. Find the perimeter and the area of each shape. All corners are right angles.

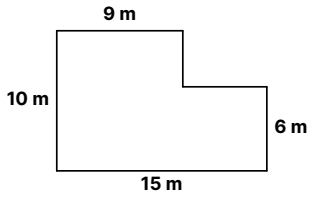
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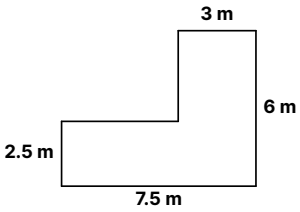
b)



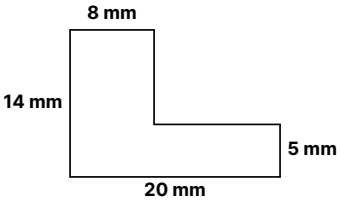
c)



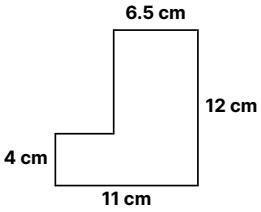
d)



e)



f)

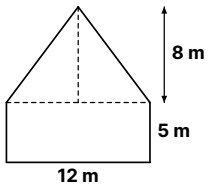


B Shapes with triangles ●●○ WORK IN YOUR BOOK

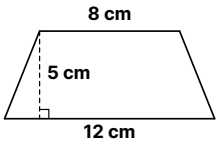
Remember: Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$. Area of a trapezium $= \frac{1}{2}(a+b) \times h$.

2. Find the area of each shape.

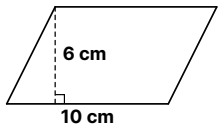
a)



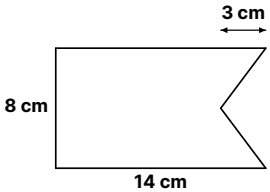
b)



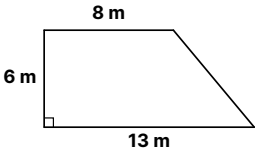
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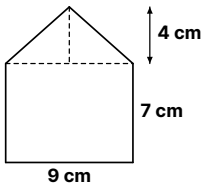
d)



e)



f)



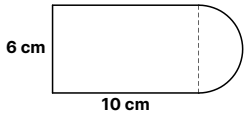
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C Shapes with circles ●●● WORK IN YOUR BOOK

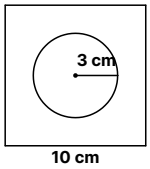
Remember: Area of a circle $= \pi r^2$. Circumference $= \pi d$. Give answers to 1 decimal place.

3. Find the area of each shape. In (b) the circle has been cut out. In (d) find the area inside the square but outside the circle.

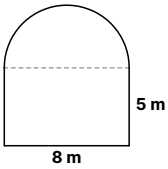
a)



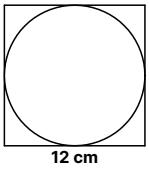
b)



c)



d)



4. Now find the perimeter of:

a) shape (a)

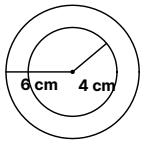
b) shape (c)

D Shaded regions ●●● WORK IN YOUR BOOK

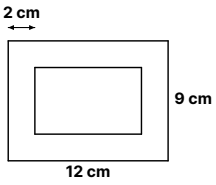
Remember: A shaded region is the big shape minus the hole. Find each area separately, then subtract.

5. Find each shaded area, to 1 decimal place where needed.

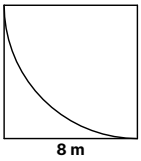
a)



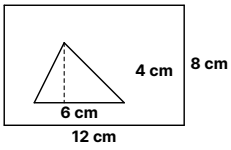
b)



c)



d)



E Problems ●●● WORK IN YOUR BOOK

6. A rectangular lawn is 12 m long and 8 m wide. A circular flower bed with a radius of 2 m is cut out of it.

a) Find the area of the grass, to 1 decimal place.

b) Fertiliser for the grass costs $3.50 for each square metre. Find the cost of fertilising the lawn.

7. A wall is 6 m long and 2.5 m high. It has a window 1.2 m by 1.5 m and a door 0.8 m by 2 m. Find the area of wall to be painted.

8. A running track has two straight sides, each 100 m long, joined by a semicircle at each end. The semicircles have a diameter of 60 m. Find the distance once around the track, to the nearest metre.

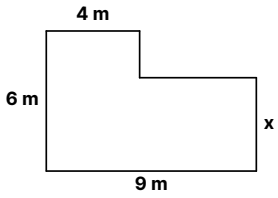
9. A round table top with a diameter of 1.2 m is cut from a square board with sides of 1.2 m. Find the area of board that is wasted, to 2 decimal places.

★ Extension FOR EARLY FINISHERS

10. A square has sides of 10 cm. A quarter circle with a radius of 5 cm is cut from each of its four corners. Find the area that is left, to 1 decimal place.

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11. The L-shape has an area of 44 m². Find the length marked x.



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Compound 2D Shapes

Name _____ Date _____ Calculator allowed Version 2

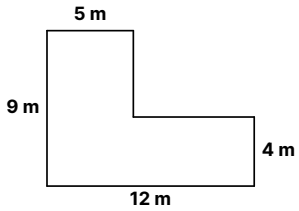
KEY WORDS composite shape · perimeter · area · rectangle · triangle · trapezium · semicircle

A Shapes made from rectangles ●○○ WORK IN YOUR BOOK

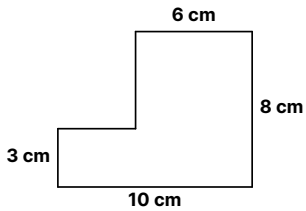
Remember: Find the missing side lengths first. For the area, split the shape into rectangles.

1. Find the perimeter and the area of each shape. All corners are right angles.

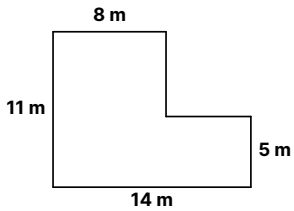
a)



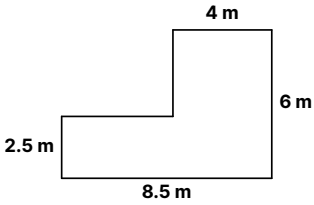
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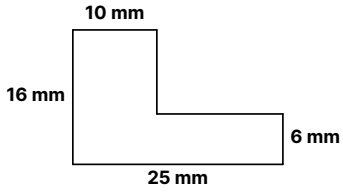
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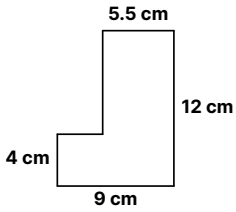
d)



e)



f)

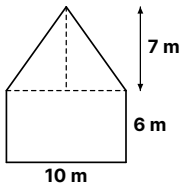


B Shapes with triangles ●●○ WORK IN YOUR BOOK

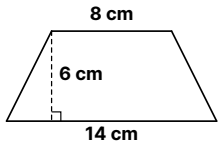
Remember: Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$. Area of a trapezium $= \frac{1}{2}(a + b) \times h$.

2. Find the area of each shape.

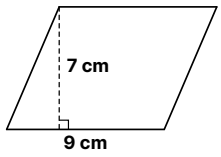
a)



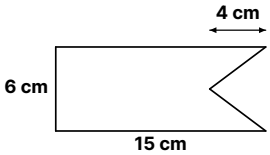
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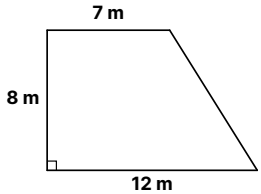
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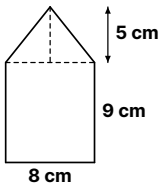
d)



e)



f)



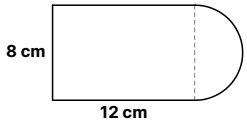
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C Shapes with circles ●●● WORK IN YOUR BOOK

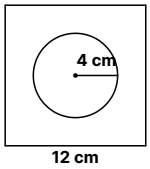
Remember: Area of a circle $= \pi r^2$. Circumference $= \pi d$. Give answers to 1 decimal place.

3. Find the area of each shape. In (b) the circle has been cut out. In (d) find the area inside the square but outside the circle.

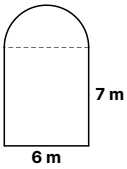
a)



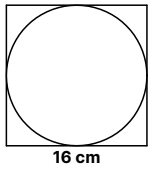
b)



c)



d)



4. Now find the perimeter of:

a) shape (a)

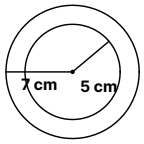
b) shape (c)

D Shaded regions ●●● WORK IN YOUR BOOK

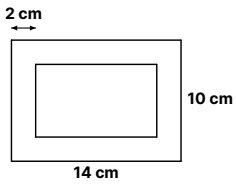
Remember: A shaded region is the big shape minus the hole. Find each area separately, then subtract.

5. Find each shaded area, to 1 decimal place where needed.

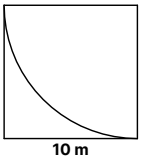
a)



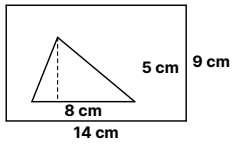
b)



c)



d)



E Problems ●●● WORK IN YOUR BOOK

6. A rectangular lawn is 15 m long and 9 m wide. A circular flower bed with a radius of 3 m is cut out of it.

a) Find the area of the grass, to 1 decimal place.

b) Fertiliser for the grass costs $4.50 for each square metre. Find the cost of fertilising the lawn.

7. A wall is 7 m long and 2.4 m high. It has a window 1.5 m by 1.2 m and a door 0.9 m by 2 m. Find the area of wall to be painted.

8. A running track has two straight sides, each 80 m long, joined by a semicircle at each end. The semicircles have a diameter of 50 m. Find the distance once around the track, to the nearest metre.

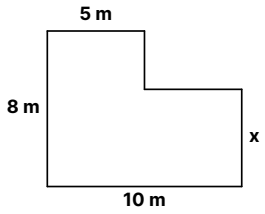
9. A round table top with a diameter of 1.4 m is cut from a square board with sides of 1.4 m. Find the area of board that is wasted, to 2 decimal places.

★ Extension FOR EARLY FINISHERS

10. A square has sides of 12 cm. A quarter circle with a radius of 6 cm is cut from each of its four corners. Find the area that is left, to 1 decimal place.

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11. The L-shape has an area of 65 m². Find the length marked x.



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